

Algebra II with Trigonometry

Chapter 7.2

Mental Math (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4582337_alg-2trig-h-chp-7-2-note-docx.pdf](#)

1. **Exact Tangent** Find the exact value of $\tan 165^\circ$.
2. **Exact Cosine in Radians** Find the exact value of $\cos\left(\frac{17\pi}{12}\right)$.
3. **Angle Difference with Sine** Find the exact value of $\sin\left(-\frac{5\pi}{12}\right)$.
4. **Recognize a Formula** Evaluate: $\cos 67^\circ \cos 23^\circ - \sin 67^\circ \sin 23^\circ$.
5. **Tangent Subtraction** Evaluate: $\frac{\tan 55^\circ - \tan 10^\circ}{1 + \tan 55^\circ \tan 10^\circ}$.
6. **Inverse Trig as Angles** Find the exact value of $\cos(\sin^{-1}(x) - \tan^{-1}(y))$ in terms of x and y .
7. **Mixed Inverse Trig** Find the exact value of $\sin\left(\cos^{-1}\frac{1}{2} + \tan^{-1}(1)\right)$.
8. **Quadrant Thinking** If $\sin \theta = \frac{4}{5}$ with θ in Quadrant II, and $\tan \phi = \frac{1}{3}$ with ϕ in Quadrant III, find $\sin(\theta - \phi)$.
9. **Single Sine Form** Rewrite $-3 \sin x + \cos x$ in the form $k \sin(x + \phi)$.
10. **Identity Insight** Simplify mentally:

$$\frac{\sin(x + y) - \sin(x - y)}{\cos(x + y) + \cos(x - y)}.$$